

Lecture Note in Calculus B - 1st

Dr. Fares Al-Azemi

Done by: Haneen Alfadhli
@HAlfadhli17

4.4-Indeterminate form and l'Hopital's rule

L'Hospital's Rule: Suppose f and g are differentiable and $g'(x) \neq 0$ on an open Interval I that contains a (except possibly at a), if

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ has the form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ then ,

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, if the limit on the right side exist or $\pm \infty$.

CASES:

- $\frac{0}{0}$ or $\frac{\infty}{\infty} \rightarrow L.R$
- $0.(\infty)$ or $0.(-\infty) \rightarrow f.g = \frac{f}{1/g}$ or $\frac{g}{1/f} \rightarrow L.R$
- $1^\infty, 0^0, \infty^0 \rightarrow$ taking natural logarithm

Examples

1. $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \frac{0}{0}$

Sol:

$$\rightarrow \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} = 4$$

$$\text{or by L.R : } \lim_{x \rightarrow 2} \frac{2x}{1} = 4$$

$$2. \lim_{x \rightarrow 0} \frac{-1-x+e^x}{x^2} = \frac{0}{0}$$

Sol:

$$\rightarrow L.R : \lim_{x \rightarrow 0} \frac{-1 + e^x}{2x} = \frac{0}{0}$$

$$\rightarrow L.R \text{ again : } \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

$$3. \lim_{x \rightarrow \frac{\pi^-}{2}} (2x - \pi) \sec x = \lim_{x \rightarrow \frac{\pi^-}{2}} \frac{2x - \pi}{\cos x} = \frac{0}{0}$$

Sol:

$$\rightarrow L.R : \lim_{x \rightarrow \frac{\pi^-}{2}} \frac{2}{-\sin x} = \frac{2}{-1} = -2$$

$$4. \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} = \frac{\infty}{\infty}$$

Sol:

$$\rightarrow L.R : \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{3} x^{-2/3}} = 3 \lim_{x \rightarrow \infty} \frac{x^{2/3}}{x} = 3 \lim_{x \rightarrow \infty} \frac{1}{\sqrt[3]{x}} = 0$$

$$5. \lim_{x \rightarrow 0^+} \left[\frac{1}{e^x - 1} - \frac{1}{x} \right] = \infty - \infty$$

Sol:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{x - e^x + 1}{x(e^x - 1)} &= \frac{0}{0} \\ \rightarrow L.R : \lim_{x \rightarrow 0^+} \frac{1 - e^x}{e^x - 1 + xe^x} &= \frac{0}{0} \\ \rightarrow L.R : \lim_{x \rightarrow 0^+} \frac{-e^x}{e^x(x + 2)} &= \frac{-1}{2} \end{aligned}$$

$$6. \lim_{x \rightarrow 0^+} x \ln x = 0. (-\infty)$$

Sol:

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = \frac{-\infty}{\infty} \\ \rightarrow L.R : \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} &= \lim_{x \rightarrow 0^+} -x = 0 \end{aligned}$$

$$7. \lim_{x \rightarrow 0^+} x^x = 0^0$$

Sol:

$$\text{let } y = x^x$$

$$\ln y = x \ln x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}$$

$$\rightarrow L.R : \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = 0 \quad \rightarrow \lim_{x \rightarrow 0^+} \ln y = 0$$

$$\lim_{x \rightarrow 0^+} e^{\ln y} = e^0 \rightarrow \lim_{x \rightarrow 0^+} y = 1$$

$$8. \lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x}\right)^x = \infty^0, \text{ let } y = \left(1 + \frac{1}{x}\right)^x \rightarrow \ln y = x \ln \left(1 + \frac{1}{x}\right)$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln \left(1 + \frac{1}{x}\right) = 0 \cdot \infty$$

$$\lim_{x \rightarrow 0^+} \frac{\ln\left(1 + \frac{1}{x}\right)}{\frac{1}{x}} = \frac{\infty}{\infty}$$

$$\rightarrow L.R: \lim_{x \rightarrow 0^+} \frac{\frac{1}{\left(1 + \frac{1}{x}\right)} \cdot \left(\frac{-1}{x^2}\right)}{\frac{-1}{x^2}} = 0 \rightarrow \lim_{x \rightarrow 0^+} e^{\ln y} = e^0$$

$$\rightarrow \lim_{x \rightarrow 0^+} y = 1$$

$$9. \lim_{x \rightarrow \infty} \frac{2^x + 4^x}{5^x + 2^x} = \frac{\infty}{\infty} \rightarrow \lim_{x \rightarrow \infty} \frac{\frac{2^x}{5^x} + \frac{4^x}{5^x}}{1 + \frac{2^x}{5^x}} = \frac{0+0}{1+0} = \frac{0}{1} = 0$$

$$10. \lim_{x \rightarrow \infty} \left[\frac{1 + \cos x}{x} \right] = \lim_{x \rightarrow \infty} \left[\frac{1}{x} + \frac{\cos x}{x} \right]$$

Sol:

$$-1 \leq \cos x \leq 1$$

$$\frac{-1}{x} \leq \frac{\cos x}{x} \leq \frac{1}{x}$$

$$\lim_{x \rightarrow \infty} \frac{-1}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\lim_{x \rightarrow \infty} \frac{\cos x}{x} = 0, (by \text{ squeeze Thm.}) \rightarrow \lim_{x \rightarrow \infty} \left[\frac{1}{x} + \frac{\cos x}{x} \right] = 0$$

11. If $f'(x)$ is continuous use L'Hospital's Rule to show that :

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h} = f'(x)$$

Sol:

Exercises-4.4

$$1. \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} \rightarrow (\text{ans.} = 3)$$

$$2. \lim_{x \rightarrow \infty} \frac{\ln x}{x} \rightarrow (\text{ans.} = 0)$$

$$3. \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \rightarrow (\text{ans.} = 1)$$

$$4. \lim_{x \rightarrow 0} \frac{\sin x}{x} \rightarrow (\text{ans.} = 1)$$

$$5. \lim_{x \rightarrow 0^+} x \ln x \rightarrow (\text{ans.} = 0)$$

$$6. \lim_{x \rightarrow 0^+} \sin^{-1}(x) \cdot \ln x \rightarrow (\text{ans.} = 0)$$

$$7. \lim_{x \rightarrow \infty} (\tanh x)^x \rightarrow (\text{ans.} = 1)$$

$$8. \lim_{x \rightarrow 1^+} \left(\frac{4}{\pi} \tan^{-1} x \right)^{\frac{1}{\ln x}} \rightarrow (\text{ans.} = e^{\frac{2}{\pi}})$$

$$9. \lim_{x \rightarrow \infty} \frac{\ln(7x+9)}{\ln(10x+100)} \rightarrow (\text{ans.} = 1)$$

$$10. \lim_{x \rightarrow \infty} x \sin \frac{1}{x} \rightarrow (\text{ans.} = 1)$$

$$11. \lim_{x \rightarrow \infty} \frac{x + \cos x}{x} \rightarrow (\text{ans.} = 1)$$

$$12. \lim_{x \rightarrow 0^+} \left[\frac{1}{x} - \frac{1}{\sin x} \right] \rightarrow (\text{ans.} = 0)$$

$$13. \lim_{x \rightarrow \infty} (\sqrt{x+1} - \sqrt{x}) \rightarrow (\text{ans.} = 0)$$

$$14. \lim_{x \rightarrow \infty} \ln(2x+1) - \ln(x+7) \rightarrow (\text{ans.} = \ln 2)$$

$$15. \text{ show that } \lim_{x \rightarrow 0+} \left(\frac{5^x + 7^x}{2} \right)^{1/x} \rightarrow (\text{ans.} = \sqrt{35})$$

$$16. \lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} \right)^{2x+1} \rightarrow (\text{ans.} = e^{-4})$$

$$17. \lim_{x \rightarrow \infty} \left(\frac{x+1}{x-1} \right)^x \rightarrow (\text{ans.} = e^2)$$

$$18. \lim_{x \rightarrow 0} \left(\frac{e^x}{\sin x} - \frac{1}{x} \right) \rightarrow (\text{ans.} = 1)$$

$$19. \lim_{x \rightarrow \infty} \left(\frac{x+3}{\cosh x} \right)^{1/x} \rightarrow (\text{ans.} = e^{-1})$$

$$20. \lim_{x \rightarrow 0+} \frac{x \sinh x}{1 - \cosh x} \rightarrow (\text{ans.} = -2)$$

21. for what value of a is the following equation

$$\text{true : } \lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x = e \rightarrow (\text{ans. } a = \frac{1}{2})$$

$$22. \lim_{x \rightarrow 0+} (1+x)^{1/x} \rightarrow (\text{ans.} = e)$$

$$23. \lim_{x \rightarrow 1} \left[\frac{1}{x-1} - \frac{1}{\ln x} \right] \rightarrow (\text{ans.} = \frac{-1}{2})$$

$$24. \lim_{x \rightarrow \infty} \frac{\sin x}{x} \rightarrow (\text{ans.} = 0)$$

7.1-Integration by parts

Formula for I.B.P:

$$\int u \, dv = u \, v - \int v \, du$$

OR

$$\int_a^b u \, dv = u \, v \Big|_a^b - \int_a^b v \, du$$

Proof of (I.B.P):

$$(u \, v)' = u \, v' + u' \, v$$

$$= u \, dv + v \, du$$

$$u \, v = \int u \, dv + \int v \, du$$

$$\therefore \int u \, dv = u \, v - \int v \, du$$

CASES:

- $\int x^n \left(\begin{matrix} \sin x \\ \cos x \\ \tan x \end{matrix} \middle| \begin{matrix} a^x \\ e^x \end{matrix} \right) dx \rightarrow u = x^n$
- $\int x^n \left(\begin{matrix} \sin^{-1} x \\ \cos^{-1} x \\ \tan^{-1} x \end{matrix} \middle| \begin{matrix} \ln x \\ \log x \end{matrix} \right) dx \rightarrow dv = x^n$
- $\int \left(\begin{matrix} \sin x \\ \cos x \\ \tan x \end{matrix} \middle| \begin{matrix} a^x \\ e^x \end{matrix} \right) \cdot \left(\begin{matrix} \sin^{-1} x \\ \cos^{-1} x \\ \tan^{-1} x \end{matrix} \middle| \begin{matrix} \ln x \\ \log x \end{matrix} \right) \rightarrow \text{repeated integral}$

Examples

1. $\int \ln x \, dx$

Sol:

$$u = \ln x \quad dv = dx$$

$$du = \frac{1}{x} dx \quad v = x$$

$$\begin{aligned} I &= \int \ln x \, dx = x \ln x - \int x \frac{1}{x} dx \\ &= x \ln x - \int dx \\ &= x \ln x - x + c \end{aligned}$$

2. $\int x \ln x \, dx$

Sol:

$$u = \ln x \quad dv = x \, dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^2}{2}$$

$$\begin{aligned} I &= \int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int \frac{x^2}{x} dx \\ &= \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx \\ &= \frac{x^2}{2} \ln x - \frac{1}{2} \cdot \frac{1}{2} x^2 + c \end{aligned}$$

3. $\int \sin^{-1} x \, dx$

Sol:

$$u = \sin^{-1} x \quad dv = dx$$

$$du = \frac{1}{\sqrt{1-x^2}} dx \quad v = x$$

$$\begin{aligned} I &= \int \sin^{-1} x \, dx = x \sin^{-1} x - \left(-\frac{1}{2}\right) \int \frac{(-2)x}{\sqrt{1-x^2}} dx \\ &= x \sin^{-1} x + \frac{1}{2} \cdot 2\sqrt{1-x^2} + c \end{aligned}$$

4. $\int t^2 e^t \, dt$

Sol:

$$u = t^2 \quad dv = e^t dt$$

$$du = 2t \, dt \quad v = e^t$$

$$I = t^2 e^t - 2 \int t e^t dt$$

$$u = t \quad dv = e^t dt$$

$$du = dt \quad v = e^t$$

$$\begin{aligned} I &= t^2 e^t - 2 \left[t e^t - \int e^t dt \right] \\ &= t^2 e^t - 2 t e^t + 2 e^t + c \end{aligned}$$

$$5. \int_0^1 \tan^{-1} x \, dx$$

sol :

$$u = \tan^{-1} x \quad dv = dx$$

$$du = \frac{1}{1+x^2} dx \quad v = x$$

$$I = (x \tan^{-1} x) \Big|_0^1 - \left(\frac{1}{2} \right) \int_0^1 \frac{(2)x}{1+x^2} dx$$

$$\text{let } t(x) = 1 + x^2, \quad dt = 2x \, dx$$

$$t(1) = 1 + 1 = 2$$

$$t(0) = 1 + 0 = 1$$

$$\int_1^2 \frac{1}{t} dt = \ln|t| \Big|_1^2 = \ln|2| - \ln|1| = \ln|2|$$

$$I = \frac{\pi}{4} - \frac{1}{2} (\ln|2|) = \frac{\pi}{4} - \frac{\ln|2|}{2}$$

$$6. \int e^x \sin x \, dx$$

sol :

$$u = \sin x \quad dv = e^x dx$$

$$du = \cos x \, dx \quad v = e^x$$

$$I = e^x \sin x - \int e^x \cos x \, dx$$

$$u = \cos x \quad dv = e^x dx$$

$$du = -\sin x \, dx \quad v = e^x$$

$$I = e^x \sin x - [e^x \cos x + \int e^x \sin x \, dx]$$

$$I = e^x \sin x - e^x \cos x - I$$

$$2I = e^x \sin x - e^x \cos x \rightarrow I = \frac{1}{2} e^x [\sin x - \cos x] + c$$

7. a- Prove the reduction formula :

$$\int \sin^n x \, dx = \frac{-1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

Proof:

$$LHS: \int (\sin x)^n dx = \int (\sin x)^{n-1} \sin x \, dx$$

$$u = (\sin x)^{n-1} \quad dv = \sin x \, dx$$

$$du = (n-1)(\sin x)^{n-2} dx \quad v = -\cos x$$

$$I = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$$

$$= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx$$

$$= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) I$$

$$I + (n-1)I = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx$$

$$I = \frac{-1}{n} \cos x \sin^{n-1} x + \frac{(n-1)}{n} \int \sin^{n-2} x \, dx = RHS$$

b- $\int_0^{\pi/2} \sin^4 x \, dx \quad : n \geq 2$

sol :

$$\begin{aligned}
 &= \left. \frac{-1}{4} \cos x \sin^3 x \right|_0^{\pi/2} + \frac{3}{4} \int_0^{\pi/2} \sin^2 x \, dx \\
 &= 0 + \frac{3}{4} \left[\int_0^{\pi/2} \frac{1}{2} (1 - \cos 2x) \, dx \right] \\
 &= \frac{3}{4} \cdot \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2} \\
 &= \frac{3}{8} \left(\frac{\pi}{2} \right) = \frac{3\pi}{16}
 \end{aligned}$$

8. Use I.B.P to evaluate $\int f(x) \, dx$ where $f(x) = \int_1^x \frac{\sin t}{t} \, dt$

sol:

FTC 

$$\begin{aligned}
 u &= \int_1^x \frac{\sin t}{t} \, dt & dv &= dx \\
 du &= \frac{\sin x}{x} \, dx & v &= x \\
 I &= x \int_1^x \frac{\sin t}{t} \, dt - \int x \frac{\sin x}{x} \, dx \\
 &= x f(x) + \cos x + c
 \end{aligned}$$

Exercises-7.1

$$1. \int \cos(\ln x) dx \rightarrow \left(ans = \frac{1}{2} (x \cos(\ln x) + x \sin(\ln x)) + c \right)$$

$$2. \int e^x \cos x dx \rightarrow \left(ans = \frac{1}{2} e^x (\sin x + \cos x) + c \right)$$

$$3. \int x \sec^{-1} x dx \rightarrow \left(ans = \frac{1}{2} x^2 \sec^{-1} x - \frac{1}{2} (x^2 - 1)^{1/2} + c \right)$$

$$4. \int \tan^{-1} \left(\frac{1}{x} \right) dx \rightarrow \left(ans = x \tan^{-1} \left(\frac{1}{x} \right) + \frac{1}{2} \ln |x^2 + 1| + c \right)$$

$$5. \int \sin x \ln(\sin x) dx$$

$$\rightarrow \left(ans = -\cos x \ln(\sin x) + \ln |\csc x - \cot x| + \cos x + c \right)$$

$$6. \int (\ln x)^2 dx \rightarrow \left(ans = x(\ln x)^2 - 2[x \ln x - x] + c \right)$$

$$7. \int \frac{\ln(x^2 e^x)}{x^3} dx \rightarrow \left(ans = -x^{-2} \ln x - \frac{1}{2} x^{-2} - x^{-1} + c \right)$$

$$8. \int x 3^x dx \rightarrow \left(ans = \frac{x}{\ln 3} 3^x - \frac{1}{(\ln 3)^2} 3^x + c \right)$$

$$9. \int \sin x \sinh(2x) dx$$

$$\rightarrow \left(ans = \frac{1}{5} (-\cos x \sinh(2x) + 2 \sin x \cos x + c) \right)$$

$$10. \int e^x \cos(2x) dx \rightarrow \left(ans = \frac{1}{5} e^x [\cos 2x + 2 \sin 2x] + c \right)$$

$$11. \int e^{\sqrt{x}} dx \rightarrow \left(ans = 2(\sqrt{x} e^{\sqrt{x}} - e^{\sqrt{x}}) + c \right)$$

$$12. \int \sin \sqrt{1+2x} \, dx$$

$$\rightarrow (ans = -\sqrt{1+2x} \cos \sqrt{1+2x} + \sin \sqrt{1+2x} + c)$$

$$13. \int \ln(1 + \sqrt{x}) \, dx$$

$$\rightarrow (ans = (\sqrt{x})^2 \ln(1 + \sqrt{x}) - \frac{1}{2}(\sqrt{x})^2 + \sqrt{x} + \ln|1 - \sqrt{x}| + c)$$

$$14. \int x (\ln x + e^x) dx \rightarrow (ans = \frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + xe^x - e^x + c)$$

$$15. \int \frac{\cos^{-1}(\ln x)}{x} dx \rightarrow (ans = \ln x \cos^{-1}(\ln x) - \sqrt{1 - (\ln x)^2} + c)$$

$$16. \text{ Suppose } f(x) \text{ continuously diff. on } [0,1] \text{ and } f(1) = -3$$

$$\int_0^1 x \sqrt{1 - f(x)} \, dx = 5, \text{ find } \int_0^1 \frac{x^2 f'(x)}{\sqrt{1 - f(x)}} \, dx \rightarrow (ans = 16)$$

$$17. \int x^3 \sin(x^2) \, dx \rightarrow (ans = \frac{1}{2}[-x^2 \cos x^2 + \sin x^2] + c)$$

7.2-Trigonometric integrals

Remark:

$$\int \frac{\sin^n x \cos^m x}{\sinh^n x \cosh^m x} dx$$

CASES:

- if n or m odd (نفك الاس الفردي الاصغر)

$$\sin^2 x = 1 - \cos^2 x \quad \text{or} \quad \cos^2 x = 1 - \sin^2 x$$

- if n and m are even (نفك الاثنين معا), use the half angle identities :

$$\sin^2 x = \frac{1}{2} [1 - \cos(2x)] \quad \text{and} \quad \cos^2 x = \frac{1}{2} [1 + \cos(2x)]$$

Examples

1. $\int \cos^3 x \, dx$

sol :

$$= \int \cos^2 x \cos x \, dx = \int (1 - \sin^2 x) \cos x \, dx$$

$$u = \sin x \rightarrow du = \cos x \, dx$$

$$I = \int (1 - u^2) du = u - \frac{u^3}{3} + c, \quad u = \sin x$$

2. $\int \sin^5 x \cos^2 x \, dx$

sol :

$$\begin{aligned}
 &= \int (\sin^2 x)^2 \cos^2 x \sin x \, dx \\
 &= - \int (1 - \cos^2 x)^2 \cos^2 x (-\sin x \, dx) \\
 &\quad u = \cos x \rightarrow du = -\sin x \, dx \\
 I &= - \int (1 - u^2)^2 u^2 du = - \int (1 - 2u^2 + u^4) u^2 du \\
 &= - \int (u^2 - 2u^4 + u^6) du \\
 &= - \left[\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} \right] + c, \quad u = \cos x
 \end{aligned}$$

3. $\int \tan^2 x \cos^5 x \, dx$

sol :

$$\begin{aligned}
 &= \int \frac{\sin^2 x}{\cos^2 x} \cos^5 x \, dx = \int \sin^2 x \cos^3 x \, dx \\
 &= \int \sin^2 x \cos^2 x \cos x \, dx = \int \sin^2 x (1 - \sin^2 x) \cos x \, dx \\
 &\quad u = \sin x \rightarrow du = \cos x \, dx \\
 I &= \int u^2 (1 - u^2) du = \int (u^2 - u^4) du \\
 &= \frac{u^3}{3} - \frac{u^5}{5} + c, \quad u = \sin x
 \end{aligned}$$

4. $\int \sin^4 x \, dx$

sol :

$$\begin{aligned}
 &= \int (\sin^2 x)^2 \, dx = \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx \\
 &= \frac{1}{4} \int (1 - 2 \cos 2x + \cos^2 2x) \, dx \\
 &= \frac{1}{4} \int \left(1 - 2 \cos 2x + \frac{1}{2} + \frac{\cos 4x}{2} \right) \, dx \\
 &= \frac{1}{4} \left[\frac{3}{2} x - \sin 2x + \frac{1}{8} \sin 4x \right] + c
 \end{aligned}$$

5. $\int (1 - \cos x)^{5/2} \, dx$

sol :

$$\begin{aligned}
 \sin^2 x &= \frac{1}{2} (1 - \cos 2x) \rightarrow \sin^2 \left(\frac{x}{2} \right) = \frac{1}{2} (1 - \cos x) \\
 &\rightarrow 2 \sin^2 \left(\frac{x}{2} \right) = 1 - \cos x \\
 I &= \int \left(2 \sin^2 \left(\frac{x}{2} \right) \right)^{5/2} \, dx = 2^{5/2} \int \left(\sin \left(\frac{x}{2} \right) \right)^5 \, dx \\
 &= \sqrt{32} \int \left(\sin \frac{x}{2} \right)^4 \sin \left(\frac{x}{2} \right) \, dx = \sqrt{32} \int \left(1 - \cos^2 \frac{x}{2} \right)^2 \sin \frac{x}{2} \, dx \\
 &\quad u = \cos \frac{x}{2} \rightarrow du = -\frac{1}{2} \sin \frac{x}{2} \\
 I &= -2\sqrt{32} \int (1 - u^2)^2 \, du = -2\sqrt{32} \int (1 - 2u^2 + u^4) \, du
 \end{aligned}$$

$$= -2\sqrt{32} \left[u - \frac{2}{3}u^3 + \frac{u^5}{5} \right] + c \quad , \quad u = \cos \frac{x}{2}$$

Remark:

$$\int \frac{\tan^n x \sec^m x}{\cot^n x \csc^m x} dx$$

CASES:

- If n is odd (tan or cot):

$$\tan^2 x = \sec^2 x - 1, \quad \cot^2 x = \csc^2 x - 1$$

- If m is even (sec or csc):

$$\sec^2 x = \tan^2 x + 1, \quad \csc^2 x = \cot^2 x + 1$$

- If n is even and m is odd:

I.B.P

Examples

1. $\int \tan^3 x \, dx$

sol :

$$\begin{aligned} &= \int \tan x \tan^2 x \, dx = \int \tan x (\sec^2 x - 1) \, dx \\ &= \int \tan x \sec^2 x \, dx - \int \tan x \, dx \\ &= \frac{1}{2} \tan^2 x - \ln|\sec x| + c \end{aligned}$$

2. $\int \tan^3 x \sec^{5/2} x \, dx$

sol :

$$\begin{aligned}
 &= \int \tan^2 x \sec^{3/2} x \tan x \sec x \, dx \\
 &= \int (\sec^2 x - 1) \sec^{3/2} x \tan x \sec x \, dx \\
 &\quad u = \sec x \rightarrow du = \sec x \tan x \, dx \\
 &= \int (u^2 - 1) u^{3/2} \, du = \int (u^{7/2} - u^{3/2}) \, du \\
 &= \frac{2}{9} u^{9/2} - \frac{2}{5} u^{5/2} + c \\
 &= \frac{2}{9} \sec^{9/2} - \frac{2}{5} \sec^{5/2} + c
 \end{aligned}$$

3. $\int \sec^3 x \, dx$

sol :

I.B.P:

$$\begin{aligned}
 &u = \sec x \quad dv = \sec^2 x \, dx \\
 &du = \sec x \tan x \, dx \quad v = \tan x \\
 &I = \sec x \tan x - \int \sec x \tan^2 x \, dx \\
 &= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx \\
 &= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx \\
 &= \sec x \tan x - I + \ln|\sec x + \tan x| + c \\
 &2I = \sec x \tan x + \ln|\sec x + \tan x| + c
 \end{aligned}$$

$$I = \frac{1}{2} [\sec x \tan x + \ln |\sec x + \tan x| + c]$$

Remark:

$$a. \sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

$$b. \sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$c. \cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

4. $\int \sin 4x \cos 5x \, dx$

sol :

$$\begin{aligned} &= \int \frac{1}{2} [\sin(4 - 5) + \sin(4 + 5)] \\ &= \frac{1}{2} \left[\int -\sin x \, dx + \int \sin 9x \, dx \right] \\ &= \frac{1}{2} \left[\cos x - \frac{1}{9} \cos 9x \right] + c \end{aligned}$$

5. $\int \csc^3(2x) \sin^6 x \, dx$

sol :

$$\begin{aligned} &= \int \frac{1}{(\sin 2x)^3} \sin^6 x \, dx = \int \frac{\sin^6 x \, dx}{2^3 \sin^3 x \cos^3 x} \\ &= \frac{1}{8} \int \tan^3 x \, dx = \frac{1}{8} \int \tan x \tan^2 x \, dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{8} \int \tan x (\sec^2 x - 1) dx \\
&= \frac{1}{8} \left[\int \tan x \sec^2 x dx - \int \tan x dx \right] \\
&= \frac{1}{8} \left[\frac{1}{2} \tan^2 x - \ln |\sec x| + c \right]
\end{aligned}$$

6. $\int \csc^4 x \cot^6 x dx$

sol :

$$\begin{aligned}
&= \int \cot^6 x (\cot^2 x + 1) \csc^2 x dx \\
&\quad u = \cot x \rightarrow du = -\csc^2 x dx \\
&= \int u^6 (u^2 + 1) (-du) = - \int (u^8 + u^6) du \\
&\quad = -\frac{u^9}{9} - \frac{u^7}{7} + c \\
&\quad = -\frac{\cot^9 x}{9} - \frac{\cot^7 x}{7} + c
\end{aligned}$$

7. $\int \cot^{3/2} x \csc^6 x dx$

sol :

$$\begin{aligned}
&= \int \cot^{3/2} x \csc^4 x \csc^2 x dx \\
&= \int \cot^{3/2} x (1 + \cot^2 x)^2 \csc^2 x dx \\
&\quad u = \cot x \rightarrow du = -\csc^2 x dx \\
&= - \int u^{3/2} (1 + u^2)^2 du = - \int u^{3/2} (1 + 2u^2 + u^4) du \\
&\quad = - \int (u^{3/2} + 2u^{7/2} + u^{11/2}) du
\end{aligned}$$

$$\begin{aligned}
&= -\frac{2}{5}u^{5/2} - 2 \cdot \frac{2}{9}u^{9/2} - \frac{2}{13}u^{13/2} + c \\
&= -\frac{2}{5}\cot^{5/2}x - 2 \cdot \frac{2}{9}\cot^{9/2}x - \frac{2}{13}\cot^{13/2}x + c
\end{aligned}$$

Exercises-7.2

$$1. \int \sin^4 x \cos^3 x \, dx \rightarrow \left(ans = \frac{1}{5} (\sin x)^5 - \frac{1}{7} (\sin x)^7 + c \right)$$

$$2. \int \frac{\sin^3(2x)}{\sec^4 x} \, dx$$

$$\rightarrow \left(ans = -8 \left[\frac{1}{8} (\cos x)^8 - \frac{1}{10} (\cos x)^{10} \right] + c \right)$$

$$3. \int \sin^3 x \cos\left(\frac{x}{2}\right) \, dx$$

$$\rightarrow \left(ans = -16 \left[\frac{1}{5} (\cos x)^5 - \frac{1}{7} (\cos x)^7 \right] + c \right)$$

$$4. \int (\cos x)^{-1/2} \sin^3 x \, dx$$

$$\rightarrow \left(ans = -2\sqrt{\cos x} + \frac{2}{5} (\cos x)^{5/2} + c \right)$$

$$5. \int \sin^2 x \cos^2 x \, dx \rightarrow \left(ans = \frac{1}{8} \left[x - \frac{\sin 4x}{4} \right] + c \right)$$

$$6. \int \sin^4 x \, dx \rightarrow \left(ans = \frac{1}{4} \left[x - \frac{2 \sin 2x}{2} + \frac{x}{2} + \frac{1}{2} \frac{\sin 4x}{4} \right] + c \right)$$

$$7. \int \cos^5 x \, dx \rightarrow \left(ans = \sin x - \frac{2}{3} (\sin x)^3 + \frac{1}{5} (\sin x)^5 + c \right)$$

$$8. \int_0^{\pi} \sin x \cos x \cos(3x) \, dx \rightarrow \left(ans = \frac{-2}{5} \right)$$

$$9. \int \sin(3x) \sin(5x) \, dx \rightarrow \left(ans = \frac{1}{2} \left[\frac{\sin 2x}{2} - \frac{\sin 8x}{8} \right] + c \right)$$

$$10. \int \cos(3x) \cos^2(4x) \, dx$$

$$\rightarrow \left(ans = \frac{1}{2} \left[\frac{\sin 3x}{3} \right] + \frac{1}{4} \left[\frac{\sin 11x}{11} + \frac{\sin 5x}{5} \right] + c \right)$$

$$11. \int \frac{\csc^4 x}{\sqrt[3]{\cot^2 x}} \, dx \rightarrow \left(ans = - \left[3(\cot x)^{1/3} + \frac{3}{7} (\cot x)^{7/3} \right] + c \right)$$

$$12. \int \operatorname{sech}^6 x \, dx$$

$$\rightarrow \left(\text{ans} = \tanh x - \frac{2}{3} (\tanh x)^3 + \frac{1}{5} (\tanh x)^5 + c \right)$$

$$13. \int \tanh^3 x \, dx \rightarrow \left(\text{ans} = \ln |\cosh x| + \frac{(\cosh x)^{-2}}{2} + c \right)$$

$$14. \int \tan^2 x \sec x \, dx \rightarrow \left(\text{ans} = \sec x \tan x - \frac{1}{2} \ln |\sec x + \tan x| + c \right)$$

$$15. \int \frac{\tan^3 x \sec^4 x + \sin x}{\sec^3 x} \, dx$$

$$\rightarrow \left(\text{ans} = \frac{1}{3} (\sec x)^3 - \sec x - \frac{1}{4} (\cos x)^4 + c \right)$$

$$16. \int (\sin x)^{-1/2} (\cos x)^{-3/2} \, dx \rightarrow \left(\text{ans} = 2(\tan x)^{1/2} + c \right)$$

$$17. \int \csc^3 x$$

$$18. \int \cot^2 x \csc x \, dx$$

$$19. \int \tan^2 x \sec^3 x \, dx$$

7.3-Trigonometric substitutions

Remark:

We can transform the integral into a function for which trigonometric substitution can be used.

Use this substitution	in this expression	... here's the arithmetic	simplified radical	domain of t
$x = a \sin(t)$	$\sqrt{a^2 - x^2}$	$\sqrt{a^2 - a^2 \sin^2 t} = \sqrt{a^2(1 - \sin^2 t)} = \sqrt{a^2 \cos^2 t}$	$= a \cos(t)$	$t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$x = a \tan(t)$	$\sqrt{a^2 + x^2}$	$\sqrt{a^2 + a^2 \tan^2 t} = \sqrt{a^2(1 + \tan^2 t)} = \sqrt{a^2 \sec^2 t}$	$= a \sec(t)$	$t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$x = a \sec(t)$	$\sqrt{x^2 - a^2}$	$\sqrt{a^2 \sec^2 t - a^2} = \sqrt{a^2(\sec^2 t - 1)} = \sqrt{a^2 \tan^2 t}$	$= a \tan(t)$	$t \in [0, 2\pi]$

Examples

1. $\int \frac{\sqrt{9-x^2}}{x^2} dx$

$$\sqrt{9-x^2} = \sqrt{3^2-x^2}$$

$$\text{let } x = 3 \sin \theta \rightarrow dx = 3 \cos \theta d\theta$$

$$\sin \theta = \frac{x}{3} \rightarrow \theta = \sin^{-1} \left(\frac{x}{3} \right)$$

$$\sqrt{9-x^2} = \sqrt{9-9 \sin^2 \theta} = \sqrt{9(1-\sin^2 \theta)} = 3 \cos \theta$$

$$\begin{aligned} I &= \int \frac{(3 \cos \theta) \cdot (3 \cos \theta d\theta)}{9 \sin^2 \theta} = \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \cot^2 \theta d\theta \\ &= \int (\csc^2 \theta - 1) d\theta = -\cot \theta - \theta + c \end{aligned}$$

$$= -\frac{\sqrt{9-x^2}}{x} - \sin^{-1} \left(\frac{x}{3} \right) + c$$

2. $\int \frac{dx}{\sqrt{4+x^2}}$

$$\sqrt{4+x^2} = \sqrt{2^2+x^2}$$

$$\text{let } x = 2 \tan \theta \rightarrow dx = 2 \sec^2 \theta d\theta$$

$$\tan \theta = \frac{x}{2} \rightarrow \theta = \tan^{-1} \left(\frac{x}{2} \right)$$

$$\sqrt{4+x^2} = \sqrt{4+4 \tan^2 \theta} = \sqrt{4(1+\tan^2 \theta)} = 2 \sec \theta$$

$$I = \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + c$$

$$= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + c$$

$$3. \int \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

$$\sqrt{x^2 - 1} = \sqrt{x^2 - 1^2}$$

$$\text{let } x = \sec \theta \rightarrow dx = \sec \theta \tan \theta d\theta \rightarrow \theta = \sec^{-1} x$$

$$\sqrt{x^2 - 1} = \sqrt{\sec^2 \theta - 1} = \sqrt{\tan^2 \theta} = \tan \theta$$

$$\begin{aligned} I &= \int \frac{\sec \theta \tan \theta d\theta}{\sec^2 \theta \tan \theta} = \int \frac{1}{\sec \theta} d\theta = \int \cos \theta d\theta \\ &= \sin \theta + c = \frac{\sqrt{x^2 - 1}}{x} + c \end{aligned}$$

Remark:

Completing the square formula:

$$ax^2 + bx = (cx + d)^2 + e$$

$$c = \sqrt{a} \quad , \quad d = \frac{b}{2\sqrt{a}} \quad , \quad e = -(d)^2$$

$$4. \int \frac{dx}{\sqrt{x^2 + 8x + 25}}$$

$$x^2 + 8x + 25 = (x + 4)^2 + 9 \rightarrow I = \int \frac{dx}{\sqrt{(x+4)^2 + 3^2}}$$

$$\text{let } x + 4 = 3 \tan \theta \rightarrow dx = 3 \sec^2 \theta d\theta$$

$$\begin{aligned} I &= \int \frac{3 \sec^2 \theta d\theta}{3 \sec \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + c \\ &= \ln \left| \frac{\sqrt{(x+4)^2 + 9}}{3} + \frac{x+4}{3} \right| + c \end{aligned}$$

$$5. \int \frac{x^3}{(4x^2+9)^{3/2}} dx$$

$$4x^2 + 9 = (2x)^2 + 3^2$$

$$2x = 3 \tan \theta \rightarrow x = \frac{3}{2} \tan \theta \rightarrow dx = \frac{3}{2} \sec^2 \theta d\theta$$

$$I = \int \frac{\left(\frac{3}{2} \tan \theta\right)^3 \cdot \left(\frac{3}{2} \sec^2 \theta d\theta\right)}{\left(\sqrt{4 \cdot \frac{9}{4} \tan^2 \theta + 9}\right)^3} = \frac{3}{2^4} \int \tan^3 \theta \frac{1}{\sec \theta} d\theta$$

$$= \frac{3}{16} \int \frac{\sin^3 \theta}{\cos^3 \theta} \cdot \cos \theta d\theta = \frac{3}{16} \int \sin^3 \theta \cos^{-2} \theta d\theta$$

$$= \frac{3}{16} \int (1 - \cos^2 \theta) \cos^{-2} \theta \sin \theta d\theta$$

$$\text{let } u = \cos \theta \rightarrow du = -\sin \theta d\theta$$

$$I = \frac{-3}{16} \int (1 - u^2) u^{-2} du = \frac{3}{16} \int (1 - u^{-2}) du$$

$$= \frac{3}{16} \left(u + \frac{1}{u} \right) + c = \frac{3}{16} \left(\cos \theta + \frac{1}{\cos \theta} \right) + c$$

$$= \frac{3}{16} \left(\frac{3}{\sqrt{4x^2 + 9}} + \frac{\sqrt{4x^2 + 9}}{3} \right) + c$$

Exercises-7.3

$$1. \int \frac{dx}{x^2\sqrt{4-x^2}} \rightarrow \left(ans = \frac{-1}{4} \cot \left(\sin^{-1} \left(\frac{x}{2} \right) \right) + c \right)$$

$$2. \int \frac{x^2}{\sqrt{16-x^2}} dx \rightarrow \left(ans = 8 \sin^{-1} \left(\frac{x}{4} \right) - \frac{8x}{4} \frac{\sqrt{16-x^2}}{4} + c \right)$$

$$3. \int \frac{\sqrt{x^2-25}}{x} dx \rightarrow \left(ans = 5\sqrt{x^2-25} - 5 \sec^{-1} \left(\frac{x}{5} \right) + c \right)$$

$$4. \int \frac{dx}{(x^2+36)^2} \rightarrow \left(ans = \frac{1}{432} \left(\tan^{-1} \left(\frac{x}{6} \right) + \frac{6x}{x^2+36} \right) + c \right)$$

$$5. \int \frac{dx}{x^2(x^2+1)^{3/2}} \rightarrow \left(ans = -\frac{\sqrt{x^2+1}}{x} - \frac{x}{\sqrt{x^2+1}} + c \right)$$

$$6. \int \frac{dx}{x^2-2x+5} \rightarrow \left(ans = \frac{1}{2} \tan^{-1} \left(\frac{x-1}{2} \right) + c \right)$$

$$7. \int \frac{dx}{\sqrt{2-4x-2x^2}} \rightarrow \left(ans = \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + c \right)$$

$$8. \int \frac{e^x}{\sqrt{e^{2x}+2e^x+2}} dx \rightarrow \left(ans = \ln \left| \sqrt{(e^x+1)^2+1} + e^x + 1 \right| + c \right)$$

$$9. \int \frac{x \, dx}{x^2 - 6x + 13} \rightarrow \left(ans = \frac{1}{2} \ln |(x - 3)^2 + 4| + \frac{3}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + c \right)$$

$$10. \int \frac{dx}{\sqrt{3x - x^2}} \rightarrow \left(ans = \sin^{-1} \left(\frac{x - \frac{3}{2}}{\frac{3}{2}} \right) + c \right)$$

$$11. \int (x^2 + 4x + 3)^{-3/2} \, dx \rightarrow \left(ans = \frac{-(x+2)}{\sqrt{(x+2)^2 - 1}} + c \right)$$

$$12. \int \frac{x^2 + 6x + 9}{\sqrt{7 - 6x - x^2}} \, dx$$

$$\rightarrow \left(ans = 8 \sin^{-1} \left(\frac{x+3}{4} \right) - \frac{(x+3)}{2} \sqrt{16 - (x+3)^2} + c \right)$$

$$13. \int x \sqrt{8 - 2x - x^2} \, dx$$

$$\rightarrow \left(ans = 9 \left(\frac{\sqrt{9 - u^2}}{3} \right)^3 - \frac{9}{2} \sin^{-1} \left(\frac{u}{3} \right) - \frac{1}{2} u \sqrt{9 - u^2} + c \right)$$

$$, \quad u = (x + 1)$$

7.4-Integration of rational functions by partial fractions

Remark:

1. Assume we have: $\frac{f}{g}$

- if $\deg(f) \geq \deg(g) \rightarrow \begin{cases} \text{long division} \\ \text{factorize } (g) \end{cases}$

- if $\deg(f) < \deg(g) \rightarrow \text{factorize } (g)$

2.
$$\frac{1}{(x+1)(x+2)(x-3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x-3}$$

3.
$$\frac{1}{x^3(x+1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x+1} + \frac{E}{(x+1)^2}$$

4.
$$\frac{x}{(x^2-1)(x^2+4)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+4} + \frac{Ex+F}{(x^2+4)^2}$$

Examples

1. $\int \frac{x^3+x}{x-1} dx$

$\deg(x^3 + x) \geq \deg(x - 1) \rightarrow$ long division

$$\frac{x^3 + x}{x - 1} = (x^2 + x + 2) + \left(\frac{2}{x - 1}\right)$$

$$\begin{aligned}\int \frac{x^3 + x}{x - 1} dx &= \int (x^2 + x + 2) + \left(\frac{2}{x - 1}\right) dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln|x - 1| + c\end{aligned}$$

2. $\int \frac{x^4}{x^2-1} dx$

$\deg(x^4) \geq \deg(x^2 - 1) \rightarrow$ long division

$$\frac{x^4}{x^2 - 1} = (x^2 + 1) + \frac{1}{x^2 - 1}$$

$$\frac{1}{x^2 - 1} \rightarrow \deg(1) < \deg(x^2 - 1) \rightarrow \{\text{factorize } (x^2 - 1)\}$$

$$\rightarrow \frac{1}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1} = \frac{\left(\frac{1}{2}\right)}{x-1} + \frac{\left(\frac{-1}{2}\right)}{x+1}$$

$$\begin{aligned}\int \frac{x^4}{x^2-1} dx &= \int (x^2 + 1) + \frac{\left(\frac{1}{2}\right)}{x-1} + \frac{\left(\frac{-1}{2}\right)}{x+1} dx \\ &= \frac{x^3}{3} + x + \frac{1}{2} \ln|x - 1| - \frac{1}{2} \ln|x + 1| + c\end{aligned}$$

$$3. \int \frac{4x^2+13x-9}{x^3+2x^2-3x} dx$$

$$\frac{4x^2+13x-9}{x^3-2x^2-3x} = \frac{4x^2+13x-9}{x(x^2+2x-3)} = \frac{4x^2+13x-9}{x(x+3)(x-1)}$$

$$\frac{4x^2+13x-9}{x(x+3)(x-1)} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-1}$$

$$4x^2 + 13x - 9 = A(x+3)(x-1) + Bx(x-1) + Cx(x+3)$$

$$\rightarrow A = 3, B = -1, C = 2$$

$$\begin{aligned} I &= \int \frac{4x^2+13x-9}{x^3-2x^2-3x} dx = \int \left[\frac{3}{x} + \frac{(-1)}{x+3} + \frac{2}{x-1} \right] dx \\ &= 3 \ln|x| - \ln|x+3| + 2 \ln|x-1| + c \end{aligned}$$

$$4. \int \frac{1-x-2x^2-x^3}{x(x^2+1)^2} dx$$

$$\frac{1-x-2x^2-x^3}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$\rightarrow (1-x-2x^2-x^3) = A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x$$

$$\rightarrow A = 1, B = -1, C = -1, D = 1, E = 0$$

$$\begin{aligned} I &= \int \frac{1}{x} + \frac{(-x-1)}{x^2+1} + \frac{x}{(x^2+1)^2} dx \\ &= \ln|x| - \frac{1}{2} \ln|x^2+1| - \tan^{-1} x - \frac{1}{2} \left(\frac{1}{x^2+1} \right) + c \end{aligned}$$

Remark:

- $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$

- $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right|$

5. $\int \frac{\sqrt{x+4}}{x} dx$

let $u = \sqrt{x+4} \rightarrow u^2 = (x+4) \rightarrow x = (u^2 - 4)$

$$dx = 2u du$$

$$\int \frac{\sqrt{x+4}}{x} dx = \int \frac{u \cdot (2u du)}{(u^2-4)} = \int \frac{2u^2}{(u^2-4)} du \rightarrow L.D$$

$$I = \int 2du + \int \frac{8}{(u^2-4)} du$$

$$= 2u + \frac{8}{2(2)} \ln \left| \frac{u-2}{u+2} \right| + c, \quad u = \sqrt{x+4}$$

Exercises-7.4

1. $\int \frac{dx}{x^2-x-6} \rightarrow \left(ans = \frac{1}{5} \ln|x-3| - \frac{1}{5} \ln|x+2| + c \right)$

2. $\int \frac{x^2+2x-1}{x^3-x} dx \rightarrow (ans = \ln|x| + \ln|x-1| - \ln|x+1| + c)$

3. $\int \frac{x^2+x-2}{3x^3-x^2+3x-1} dx$
 $\rightarrow \left(ans = \frac{-7}{15} \ln|3x-1| + \frac{2}{5} \ln|x^2+1| + \frac{3}{5} \tan^{-1} x + c \right)$

4. $\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2} dx$
 $\rightarrow \left(ans = x^3 + x + \frac{1}{3} \ln|x-1| + \frac{1}{3} \ln|x+2| + c \right)$

5. $\int \frac{\cosh x}{\cosh^2 x - 10} dx$
 $\rightarrow \left(ans = \frac{1}{6} \ln|\sinh x - 3| - \frac{1}{6} \ln|\sinh x + 3| + c \right)$

6. $\int \frac{\sec^4 x}{\sec^2 x - 10} dx$
 $\rightarrow \left(ans = \tan x + \frac{10}{6} \ln|\tan x - 3| - \frac{10}{6} \ln|\tan x + 3| + c \right)$

7. $\int \frac{2x}{(x^2+1)(x-1)^2} dx \rightarrow \left(ans = \frac{-1}{x-1} - \tan^{-1} x + c \right)$

8. $\int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1} dx$
 $\rightarrow \left(ans = \frac{1}{2} x^2 + x + \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + c \right)$

$$9. \int \frac{2x^2 - 10x + 5}{x^3 - 4x^2 + 5x} dx$$

$$\rightarrow \left(ans = \ln|x| + \frac{1}{2} \ln|x - 2| - 4 \tan^{-1}(x - 2) + c \right)$$

$$10. \int \frac{2 - 2x}{(x^2 + 1)(x + 1)^2} dx$$

$$\left(ans = \frac{-1}{2} \ln|x^2 + 1| - \tan^{-1} x + \ln|x + 1| - \frac{2}{x + 1} + c \right)$$

$$11. \int \frac{x^5}{x^4 - 16} dx \rightarrow \left(ans = \frac{x^2}{2} + \ln \left| \frac{(x-2)(x+2)}{x^2+4} \right| + c \right)$$

$$12. \int \frac{e^{2x}}{e^{2x} + 3e^x + 2} dx \rightarrow (ans = 2 \ln|e^x + 2| - \ln|e^x + 1| + c)$$

$$13. \int \frac{\sqrt{x} - \sqrt[6]{x}}{\sqrt{x}(x + 4\sqrt[3]{x})} dx$$

$$\rightarrow \left(ans = \frac{3}{2} \ln \left| x^{4/6} + 4 \right| - \frac{3}{2} \tan^{-1} \left(\frac{\sqrt[3]{x}}{2} \right) + c \right)$$

$$14. \int \frac{dx}{\tan x + \sin x} \rightarrow \left(ans = \frac{1}{2} \ln \left| \tan \left(\frac{x}{2} \right) \right| - \frac{1}{4} \tan^2 \left(\frac{x}{2} \right) + c \right)$$

7.5-strategy for integration

Examples

1. $\int \frac{dx}{x(\sqrt{x} + \sqrt[6]{x})}$

sol:

$$\text{let } u = \sqrt[6]{x} \rightarrow u^6 = x, u^3 = \sqrt{x}$$

$$6u^5 du = dx$$

$$I = \int \frac{6u^5 du}{u^6(u^3 + u)} = \int \frac{6 du}{u^4 + u^2} = \int \frac{6 du}{u^2(u^2 + 1)}$$

$$\frac{6}{u^2(u^2 + 1)} = \frac{A}{u} + \frac{B}{u^2} + \frac{Cu + D}{u^2 + 1}$$

$$6 = Au(u^2 + 1) + B(u^2 + 1) + (Cu + D)u^2$$

$$A = 0, B = 6, C = 0, D = -6$$

$$I = \int \left(\frac{6}{u^2} - \frac{6}{u^2 + 1} \right) du$$

$$= \frac{-6}{u} - 6 \tan^{-1}(u) + c$$

$$= \frac{-6}{\sqrt[6]{x}} - 6 \tan^{-1}(\sqrt[6]{x}) + c$$

2. $\int \sqrt{1 + \sqrt{e^x}} dx$

sol:

$$\text{let } u = \sqrt{1 + \sqrt{e^x}} \rightarrow u^2 = 1 + \sqrt{e^x}$$

$$\sqrt{e^x} = u^2 - 1 \rightarrow e^x = (u^2 - 1)^2 \rightarrow x = 2 \ln(u^2 - 1)$$

$$dx = 2 \frac{2u du}{u^2 - 1} = \frac{4u du}{u^2 - 1}$$

$$I = \int \frac{4u^2 du}{u^2 - 1} = \int \left(4 + \frac{4}{u^2 - 1} \right) du$$

$$= 4u + 4 \frac{1}{2(1)} \ln \left| \frac{u-1}{u+1} \right| + c$$

$$= 4 \sqrt{1 + \sqrt{e^x}} + 2 \ln \left| \frac{\sqrt{1 + \sqrt{e^x}} - 1}{\sqrt{1 + \sqrt{e^x}} + 1} \right| + c$$

Remark:

Karl Weierstrass substitution:

$u = \tan\left(\frac{x}{2}\right)$	$\sin x = \frac{2u}{1 + u^2}$	$\csc x = \frac{1 + u^2}{2u}$	$\tan x = \frac{2u}{1 - u^2}$
$dx = \frac{2 du}{1 + u^2}$	$\cos x = \frac{1 - u^2}{1 + u^2}$	$\sec x = \frac{1 + u^2}{1 - u^2}$	$\cot x = \frac{1 - u^2}{2u}$

3. $\int \frac{dx}{\tan x + \sin x}$
sol :

$$\text{let } t = \tan\left(\frac{x}{2}\right)$$

$$\begin{aligned} I &= \int \frac{1}{\left(\frac{2t}{1-t^2}\right) + \left(\frac{2t}{1+t^2}\right)} \cdot \left(\frac{2 dt}{1+t^2}\right) \\ &= \int \frac{1-t^2}{2t} dt = \frac{1}{2} \left[\ln|t| - \frac{t^2}{2} \right] + c \\ &= \frac{1}{2} \left[\ln \left| \tan\left(\frac{x}{2}\right) \right| - \frac{\left(\tan\left(\frac{x}{2}\right)\right)^2}{2} \right] + c \end{aligned}$$

4. $\int \frac{dx}{3 \sin x - 4 \cos x} \rightarrow \text{let } t = \tan\left(\frac{x}{2}\right)$
sol :

$$\begin{aligned} I &= \int \frac{1}{3\left(\frac{2t}{1+t^2}\right) - 4\left(\frac{1-t^2}{1+t^2}\right)} \cdot \left(\frac{2 dt}{1+t^2}\right) \\ &= \int \frac{2 dt}{4t^2 + 6t - 4} = \int \frac{dt}{2t^2 + 3t - 2} \\ \frac{1}{2t^2 + 3t - 2} &= \frac{A}{2t-1} + \frac{B}{t+2} \rightarrow A = \frac{2}{5}, B = \frac{-1}{5} \\ I &= \int \left[\left(\frac{2}{5}\right) \cdot \frac{1}{2t-1} - \left(\frac{1}{5}\right) \cdot \frac{1}{t+2} \right] dt \\ &= \frac{1}{5} [\ln|2t-1| - \ln|t+2|] + c, \quad t = \tan\left(\frac{x}{2}\right) \end{aligned}$$

5. $\int \frac{\tan x}{\sec^2 x} dx$
sol :

$$\begin{aligned} I &= \int \frac{\sin x}{\cos x} \cdot \cos^2 x dx \\ &= \frac{1}{2} \int 2 \sin x \cos x dx \\ &= \frac{1}{2} \int \sin 2x dx = -\frac{1}{2} \frac{\cos 2x}{2} + c \end{aligned}$$

6. $\int (\sin x + \cos x)^2 dx$
sol :

$$\begin{aligned} I &= \int (\sin^2 x + 2 \sin x \cos x + \cos^2 x) dx \\ &= \int (1 + \sin 2x) dx = x - \frac{\cos 2x}{2} + c \end{aligned}$$

7. $\int \frac{dx}{1-\cos x}$
sol :

$$\begin{aligned} I &= \int \frac{dx}{1-\cos x} \cdot \frac{1+\cos x}{1+\cos x} \\ &= \int \frac{1+\cos x}{1-\cos^2 x} dx \\ &= \int \frac{1+\cos x}{\sin^2 x} dx \\ &= \int \frac{1}{\sin^2 x} dx + \int \frac{\cos x}{\sin^2 x} dx \\ &= -\cot x - \csc x + c \end{aligned}$$

8. $\int e^{\sqrt{x}} dx$

sol:

$$\text{let } t = \sqrt{x} \rightarrow t^2 = x \rightarrow dx = 2t dt$$

$$I = \int e^t \cdot 2t dt$$

$$u = 2t, \quad dv = e^t dt$$

$$du = 2dt, \quad v = e^t$$

$$I = 2te^t - 2e^t + c$$

9. $\int 2^{-x} e^{x+3} dx$

sol:

$$I = e^3 \int 2^{-x} e^x dx$$

$$= e^3 \int \frac{e^x}{2^x} dx$$

$$= e^3 \int \left(\frac{e}{2}\right)^x dx$$

$$= e^3 \frac{\left(\frac{e}{2}\right)^x}{\ln\left(\frac{e}{2}\right)} + c$$

10. $\int \sqrt{\frac{1-x}{1+x}} dx$

sol:

$$\begin{aligned} I &= \int \frac{\sqrt{1-x}}{\sqrt{1+x}} dx \cdot \frac{\sqrt{1+x}}{\sqrt{1+x}} \\ &= \int \frac{\sqrt{1-x^2}}{1+x} dx \end{aligned}$$

$$x = \sin \theta \rightarrow \theta = \sin^{-1} x \rightarrow dx = \cos \theta d\theta$$

$$= \int \frac{\cos \theta}{1 + \sin \theta} \cos \theta d\theta = \int \frac{\cos^2 \theta}{1 + \sin \theta} d\theta$$

$$= \int \frac{1 - \sin^2 \theta}{1 + \sin \theta} d\theta$$

$$= \int \frac{(1 + \sin \theta) \cdot (1 - \sin \theta)}{1 + \sin \theta} d\theta$$

$$= \int (1 - \sin \theta) d\theta$$

$$= \theta + \cos \theta + c$$

$$= \sin^{-1} x + \sqrt{1-x^2} + c$$

Exercises-7.5

$$1. \int \frac{\sin^{5/2} x}{\sec^3 x} dx \rightarrow \left(\text{ans} = \frac{2}{7} \sin^{\frac{7}{2}} x - \frac{2}{11} \sin^{\frac{11}{2}} x + c \right)$$

$$2. \int \tanh^5 x \operatorname{sech} x dx$$

$$\rightarrow \left(\text{ans} = -\operatorname{sech} x + \frac{2}{3} \operatorname{sech}^3 x - \frac{1}{5} \operatorname{sech}^5 x + c \right)$$

$$3. \int \cos(\ln x) dx \rightarrow \left(\text{ans} = \frac{1}{2} (x \cos(\ln x) + x \sin(\ln x)) + c \right)$$

$$4. \int \frac{x^{\frac{1}{2}-1}}{x \left(x^{\frac{1}{6}} + x^{\frac{1}{2}} \right)} dx$$

$$\rightarrow \left(\text{ans} = 6 \frac{1}{x^{\frac{1}{6}}} + 3 \ln |x^3 + 1| + 6 \tan^{-1} \left(x^{\frac{1}{6}} \right) + c \right)$$

$$5. \int \frac{x}{\sqrt{8+2x^2-x^4}} dx \rightarrow \left(\text{ans} = \frac{1}{2} \sin^{-1} \left(\frac{x^2-1}{3} \right) + c \right)$$

$$6. \int \frac{dx}{x\sqrt{\sqrt{x}-4}} \rightarrow \left(\text{ans} = 2 \sec^{-1} \left(\frac{\sqrt[4]{x}}{2} \right) + c \right)$$

$$7. \int \frac{\sec x}{2 \sec x + \tan x + 1} dx \rightarrow \left(\text{ans} = \sqrt{2} \tan^{-1} \left(\frac{\tan\left(\frac{x}{2}\right) + 1}{\sqrt{2}} \right) + c \right)$$

$$8. \int \cos(3x) \sin^2 x dx$$

$$\rightarrow \left(\text{ans} = \frac{1}{6} \sin(3x) - \frac{1}{20} \sin(5x) - \frac{1}{4} \sin x + c \right)$$

$$9. \int \frac{2 + \tan\left(\frac{x}{2}\right)}{2 \sin x + 2 \cos x + 3} dx \rightarrow \left(\text{ans} = 2 \ln \left| \sqrt{(u+2)^2} \right| + c \right)$$

$$10. \int \sqrt{1 - \sin x} dx \rightarrow \left(\text{ans} = 2\sqrt{1 + \sin x} + c \right)$$

7.8-Improper Integrals

Type [1]:

- If $\int_a^t f(x)dx$ exists for every number $t \geq a$, then

$$\int_a^{\infty} f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$$

Provided this limit exists (as a finite number).

- If $\int_t^b f(x)dx$ exists for every number $t \leq b$, then

$$\int_{-\infty}^b f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx$$

Provided this limit exists (as a finite number).

The improper integrals $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called **convergent** if the

Corresponding limit exists and **divergent** if the limit does not exist.

- If both $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we define

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^{\infty} f(x)dx$$

Examples

1- $\int_1^{\infty} \frac{1}{x} dx$

sol:

$$\begin{aligned} I &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\ln|x|]_1^t \\ &= \lim_{t \rightarrow \infty} [\ln|t| - \ln|1|] = \ln|\infty| = \infty \\ \therefore \int_1^{\infty} \frac{1}{x} dx &\text{ is divergent} \end{aligned}$$

2- $\int_1^{\infty} \frac{1}{x^2} dx$

sol:

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx = - \lim_{t \rightarrow \infty} \left[\frac{1}{x} \right]_1^t \\ &= - \lim_{t \rightarrow \infty} \left[\frac{1}{t} - 1 \right] = 1 \\ \therefore \int_1^{\infty} \frac{1}{x^2} dx &\text{ is convergent} \end{aligned}$$

$$3- \int_{-\infty}^0 x e^x dx$$

sol:

$$\begin{aligned} I &= \lim_{t \rightarrow -\infty} \int_t^0 x e^x dx \\ u &= x, \quad dv = e^x dx \\ du &= dx, \quad v = e^x \\ I &= \lim_{t \rightarrow -\infty} [x e^x - e^x]_t^0 \\ &= \lim_{t \rightarrow -\infty} [(0 - 1) - (t e^t - e^t)] \\ &= -1 - \lim_{t \rightarrow -\infty} t e^t + \lim_{t \rightarrow -\infty} e^t = -1 \end{aligned}$$

$$4- \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

sol:

$$\begin{aligned} I &= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx \\ I_1 &= \int_{-\infty}^0 \frac{1}{1+x^2} dx = \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{1+x^2} dx \\ &= \lim_{t \rightarrow -\infty} [\tan^{-1} x]_t^0 = \frac{\pi}{2} \\ I_2 &= \int_0^{\infty} \frac{1}{1+x^2} dx = \lim_{t \rightarrow \infty} [\tan^{-1} x]_0^{\infty} = \frac{\pi}{2} \\ I &= I_1 + I_2 = \pi \end{aligned}$$

Type [2]:

- If f is continuous on $[a, b)$ and is discontinuous at b , then

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

if this limit exists (as a finite number).

- If f is continuous on $(a, b]$ and is discontinuous at a , then

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$$

if this limit exists (as a finite number).

The improper integral $\int_a^b f(x)dx$ is called **convergent** if the corresponding limit exists and **divergent** if the limit does not exist.

- If f has a discontinuity at c , where $a < c < b$, and both $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are **convergent**, then we define

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

$$5- \int_2^5 \frac{1}{\sqrt{x-2}} dx$$

sol :

$f(x)$ has a discontinuity at $x = 2$

$$I = \lim_{t \rightarrow 2+} \int_t^5 \frac{1}{\sqrt{x-2}} dx$$

$$\text{let } u = x - 2, \quad du = dx$$

$$I = \lim_{t \rightarrow 2+} \int_{t-2}^3 \frac{1}{\sqrt{u}} du = \lim_{t \rightarrow 2+} [2\sqrt{u}]_{t-2}^3$$

$$= 2 \lim_{t \rightarrow 2+} [\sqrt{3} - \sqrt{t-2}] = 2\sqrt{3}$$

$$6- \int_0^3 \frac{1}{x-1} dx$$

sol :

f has a discontinuity at $x = 1$

$$I = \int_0^1 \frac{1}{x-1} dx + \int_1^3 \frac{1}{x-1} dx$$

$$\text{let } u = x - 1, \quad du = dx$$

$$I = \lim_{t \rightarrow 1-} \int_{-1}^{t-1} \frac{1}{u} du + \lim_{t \rightarrow 1+} \int_{t-1}^2 \frac{1}{u} du$$

$$= \lim_{t \rightarrow 1-} [\ln|u|]_{-1}^{t-1} + \lim_{t \rightarrow 1+} [\ln|u|]_{t-1}^2$$

$$= \lim_{t \rightarrow 1-} [\ln|t-1| - \ln|-1|] + \lim_{t \rightarrow 1+} [\ln|2| - \ln|t-1|]$$

$$\rightarrow \int_0^3 \frac{1}{x-1} dx \text{ is divergent}$$

$$7- \int_0^4 \frac{1}{x^2+x-6} dx$$

sol :

f has a discontinuity at $x = 2$

$$\frac{1}{x^2 + x - 6} = \frac{A}{x-2} + \frac{B}{x+3} \rightarrow A = \frac{1}{5}, B = \frac{-1}{5}$$

$$I = I_1 + I_2 = \int_0^2 \frac{1}{x^2 + x - 6} dx + \int_2^4 \frac{1}{x^2 + x - 6} dx$$

$$I_1 = \int_0^2 \frac{1}{x^2 + x - 6} dx = \lim_{t \rightarrow 2^-} \int_0^t \left[\frac{\frac{1}{5}}{x-2} + \frac{\frac{-1}{5}}{x+3} \right] dx$$

$$= \lim_{t \rightarrow 2^-} \left[\frac{1}{5} (\ln|t-2| - \ln|-2|) - \frac{1}{5} (\ln|t+3| - \ln|3|) \right]$$

I_1 is divergent

$\therefore I$ is divergent

Exercises-7.8

$$1. \int_{-\infty}^0 \frac{1}{(x-8)^{2/3}} dx \rightarrow (ans = div.)$$

$$2. \int_{-\infty}^{10} \frac{1}{(x-8)^{2/3}} dx \rightarrow (ans = div.)$$

$$3. \int_{-\infty}^{\infty} \cos^2 x dx \rightarrow (ans = div.)$$

$$4. \int_{-\infty}^{\infty} \operatorname{sech} x dx \rightarrow (ans = \pi)$$

$$5. \int_4^{\infty} \frac{x+18}{x^2+x-12} dx \rightarrow (ans = div.)$$

$$6. \int_0^1 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx \rightarrow (ans = 2e - 2)$$

$$7. \int_0^1 x \ln x dx \rightarrow (ans = \frac{-1}{4})$$

$$8. \int_{1/e}^e \frac{1}{x(\ln x)^2} dx \rightarrow (ans = div.)$$

$$9. \int_0^9 \frac{x}{\sqrt[3]{x-1}} dx \rightarrow (ans = \frac{243}{10})$$

$$10. \int_0^{\pi} \frac{\sec^2 x}{4 + \tan^2 x} dx \rightarrow (ans = \frac{\pi}{2})$$

$$11. \int_0^{\infty} x e^{-x} dx \rightarrow (ans = 1)$$